

GeoRA: Geometry-Aware Low-Rank Adaptation for RLVR

GeoRA: 基于几何感知的低秩适应方法用于 RLVR

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Abstract

摘要

Reinforcement Learning with Verifiable Rewards (RLVR) is a key paradigm for improving large-scale reasoning models. Unlike supervised fine-tuning (SFT), RLVR exhibits distinct optimization dynamics and is sensitive to the preservation of pre-trained geometric structures. However, existing parameter-efficient methods face key limitations in this regime. Low-rank adaptation methods, such as PiSSA, are primarily designed for Supervised Fine-Tuning (SFT) and do not account for the distinct optimization dynamics and geometric structures of RLVR. Conversely, directly fine-tuning the unstructured sparse parameter subspace favored by RLVR encounters efficiency bottlenecks on modern hardware. To address these challenges, we propose GeoRA (Geometry-Aware Low-Rank Adaptation), a low-rank adaptation method tailored for RLVR. Specifically, GeoRA exploits the anisotropic and compressible structure of RL update subspace, and extracts its principal directions via Singular Value Decomposition (SVD) to initialize low-rank adapters, while freezing residual components as a structural anchor during training. This design preserves the pre-trained structure and enables efficient dense computation. Experiments on Qwen and Llama models from 1.5B to 32B parameters show that GeoRA consistently outperforms strong low-rank baselines across RLVR settings in mathematics, medicine, and coding, while showing stronger generalization and less forgetting on out-of-domain tasks.

基于可验证奖励的强化学习 (RLVR) 是提升大规模推理模型的关键范式。与监督微调 (SFT) 不同, RLVR 表现出独特的优化动态, 并且对预训练几何结构的保持敏感。然而, 在这种范式下, 现有的参数高效方法面临关键限制。例如, 低秩适应方法 (如 PiSSA) 主要是为监督微调 (SFT) 设计的, 没有考虑 RLVR 独特的优化动态和几何结构。相反, 直接微调 RLVR 所偏好的无结构稀疏参数子空间在现代硬件上遇到了效率瓶颈。为了解决这些挑战, 我们提出了 GeoRA (几何感知低秩适应), 一种专门针对 RLVR 的低秩适应方法。具体而言, GeoRA 利用 RL 更新子空间的各向异性且可压缩的结构, 通过奇异值分解 (SVD) 提取其主方向以初始化低秩适配器, 同时在训练过程中将残余部分冻结作为结构锚点。这种设计保留了预训练结构, 并实现了高效的密集计算。在 Qwen 和 Llama 模型 (参数量从 1.5B 到 32B) 上的实验表明, GeoRA 在数学、医学和编程等 RLVR 设置中, 始终优于强大的低秩基线方法, 同时在领域外任务上表现出更强的泛化能力和更少的遗忘。

1 Introduction

1 引言

Large reasoning models, represented by OpenAI-o1 (OpenAI et al., 2024) and DeepSeek-R1 (DeepSeek-AI et al., 2025), have established Reinforcement Learning with Verifiable Rewards (RLVR) as a pivotal paradigm for unlocking complex reasoning capabilities. Unlike supervised fine-tuning (SFT), RLVR is better characterized as a

由 OpenAI-o1 (OpenAI 等, 2024) 和 DeepSeek-R1 (DeepSeek-AI 等, 2025) 代表的大规模推理模型, 已将可验证奖励的强化学习 (RLVR) 确立为解锁复杂推理能力的关键范式。与监督微调 (SFT) 不同,

RLVR 更好地被描述为一种

constrained optimization process (Wu et al., 2025) that amplifies latent reasoning behaviors through reward-induced sampling bias (Yue et al., 2025; Zhao et al., 2025). As a result, RLVR is particularly sensitive to update stability and its alignment with pre-trained representation geometry: overly aggressive updates can collapse behavior or degrade general capabilities. Empirically, substantial gains can emerge from modifying only a small fraction of parameters (Mukherjee et al., 2025), and recent mechanistic studies further suggest that effective RLVR updates are geometrically biased toward preserving pre-trained structure (Zhu et al., 2025; Cai et al., 2025).

受限优化过程 (Wu 等, 2025) 通过奖励诱导的采样偏差 (Yue 等, 2025; Zhao 等, 2025) 放大潜在的推理行为。因此, RLVR 对更新的稳定性及其与预训练表示几何的一致性特别敏感: 过于激进的更新可能导致行为崩溃或降低通用能力。实证研究表明, 仅修改一小部分参数即可获得显著收益 (Mukherjee 等, 2025), 而最近的机制性研究进一步表明, 有效的 RLVR 更新在几何上倾向于保留预训练结构 (Zhu 等, 2025; Cai 等, 2025)。

However, existing PEFT methods face key limitations in this regime. First, SFT-oriented low-rank adaptation methods suffer from a geometric mismatch with RLVR (Yin et al., 2025). PiSSA (Meng et al., 2025), for example, forces updates onto principal components, conflicting with RLVR’s preferred subspace while protecting core features. Second, some sparse fine-tuning methods (Mukherjee et al., 2025; Zhu et al., 2025), although more consistent with RLVR update patterns, struggle to achieve practical efficiency. Because modern hardware provides limited support for unstructured sparsity, these methods often fail to translate theoretical sparsity into real-world speedups, and may even introduce additional overhead.

然而, 现有的 PEFT 方法在此范式中面临关键限制。首先, 面向 SFT 的低秩适应方法与 RLVR 存在几何不匹配 (Yin 等, 2025)。例如, PiSSA (Meng 等, 2025) 强制更新主成分, 这与 RLVR 偏好的子空间相冲突, 同时保护核心特征。其次, 一些稀疏微调方法 (Mukherjee 等, 2025; Zhu 等, 2025), 虽然更符合 RLVR 的更新模式, 但在实现实际效率方面存在困难。由于现代硬件对无结构稀疏性的支持有限, 这些方法往往无法将理论上的稀疏性转化为实际的速度提升, 甚至可能引入额外的开销。

To address these challenges, we introduce GeoRA (Geometry-Aware Low-Rank Adaptation), a low-rank adaptation framework tailored to RLVR. Our analysis shows that the effective RLVR update subspace is anisotropic and compressible rather than isotropic, exhibiting a structured low-rank form. Based on this observation, GeoRA extracts dominant trainable directions from a geometry-constrained subspace for initialization, while keeping residual components frozen as a structural anchor. In this way, GeoRA simultaneously aligns adaptation with RLVR-specific optimization ge-

为了解决这些挑战, 我们引入了 GeoRA (几何感知的低秩适应), 这是一种专门针对 RLVR 的低秩适应框架。我们的分析表明, 有效的 RLVR 更新子空间是各向异性且可压缩的, 而不是各向同性的, 呈现出一种结构化的低秩形式。基于这一观察, GeoRA 从几何约束子空间中提取主导的可训练方向用于初始化, 同时将残余成分冻结作为结构锚点。通过这种方式, GeoRA 同时将适应与 RLVR 特定的优化几何对齐。

ometry and preserves dense matrix computation, achieving both stable optimization and hardware-efficient training. To the best of our knowledge, GeoRA is the first geometry-aware low-rank adaptation framework explicitly designed for RLVR. Our contributions are summarized as follows:

在保持几何特性并保留密集矩阵计算的同时, 实现了稳定优化和高效的硬件训练。据我们所知, GeoRA 是首个专门为 RLVR 设计的几何感知的低秩适应框架。我们的贡献总结如下:

- We propose GeoRA, a geometry-aware, low-rank, and parameter-efficient adaptation framework tailored to RLVR. By aligning low-rank adaptation with RLVR-specific optimization geometry while preserving dense computation, GeoRA overcomes the geometric mismatch of SFT-oriented low-rank methods and the efficiency bottleneck of sparse methods.
- We show that the effective RL update subspace is directional and admits a compressible

low-rank structure. GeoRA extracts dominant trainable directions via SVD within this subspace to initialize low-rank adapters, while a frozen residual component acts as a structural anchor to preserve pre-trained structure.

- Experiments on Qwen and Llama models from 1.5B to 32B parameters demonstrate that GeoRA improves training stability and consistently outperforms strong low-rank baselines across diverse RLVR settings, while showing stronger generalization and less forgetting on out-of-domain tasks.
- 我们提出 GeoRA，一个面向 RLVR 的几何感知、低秩且参数高效的适应框架。通过将低秩适应与 RLVR 特定的优化几何对齐，同时保留密集计算，GeoRA 克服了面向 SFT 的低秩方法在几何上的不匹配问题，以及稀疏方法在效率上的瓶颈。
- 我们证明有效的 RL 更新子空间是方向性的，并具有可压缩的低秩结构。GeoRA 在此子空间内通过 SVD 提取主导的可训练方向以初始化低秩适配器，而一个冻结的残差组件则作为结构锚点以保留预训练结构。
- 在 Qwen 和 Llama 模型（参数量从 1.5B 到 32B）上的实验表明，GeoRA 提高了训练稳定性，并在多种 RLVR 场景中持续优于强大的低秩基线方法，同时在领域外任务上表现出更强的泛化能力和更少的遗忘。

2 Related Work

2 相关工作

2.1 RLVR and Optimization Geometry

2.1 RLVR 与优化几何

RLVR replaces traditional reward models with deterministic verifiers (e.g., in math or coding) (Zhang et al., 2025; Lambert et al., 2025; Yuan et al., 2025). Through outcome-based feedback, it incentivizes emergent reasoning behaviors like Chain-of-Thought (CoT), establishing itself as a core paradigm for enhancing LLM reasoning (Hu et al., 2025; Liu et al., 2025b).

RLVR 用确定性验证器（例如在数学或编程中）取代传统的奖励模型（Zhang 等，2025；Lambert 等，2025；Yuan 等，2025）。通过基于结果的反馈，它激励出现链式思维（Chain-of-Thought, CoT）等涌现的推理行为，确立了其作为增强大语言模型（LLM）推理的核心范式（Hu 等，2025；Liu 等，2025b）。

Recent mechanistic analyses have delineated a sharp dichotomy between supervised fine-tuning (SFT) and reinforcement learning with verifiable rewards (RLVR). While SFT primarily injects knowledge by modifying principal weight directions (Chu et al., 2025; Jin et al., 2025b), RLVR is better characterized as a constrained optimization process (Wu et al., 2025) that amplifies latent reasoning behaviors via reward-induced sampling bias rather than introducing new capabilities (Yue et al.,

最近的机制性分析揭示了监督微调（SFT）与带有可验证奖励的强化学习（RLVR）之间的一个明显二分法。虽然 SFT 主要通过修改主权重方向来注入知识（Chu 等，2025；Jin 等，2025b），但 RLVR 更可以被描述为一个受约束的优化过程（Wu 等，2025），它通过奖励诱导的采样偏差放大潜在的推理行为，而不是引入新的能力（Yue 等，

2025; Zhao et al., 2025; Alam and Rastogi, 2025). Theoretically, these updates manifest “off the principals,” favoring low-magnitude directions orthogonal to pre-trained features (Zhu et al., 2025), consistent with the rank-1 dominance observed in early training (Cai et al., 2025). How-

ever, this regime faces stability trade-offs. KL-regularization (“RL’s Razor”) attempts to limit forgetting (Shenfeld et al., 2025) but can precipitate the “Reasoning Boundary Paradox,” where aggressive reward maximization collapses exploration diversity (Nguyen et al., 2025). Although capability degradation may be partially reversed via singular vector rotation (Jin et al., 2025a,b), adaptation remains fundamentally constrained by the “Invisible Leash,” which enforces proximity to the pre-training manifold (Wu et al., 2025). Empirically, strong gains can emerge from updating only a small fraction of parameters, suggesting that RL fine-tuning often concentrates on small subnetworks (Mukherjee et al., 2025).

2025; Zhao 等人, 2025; Alam 和 Rastogi, 2025)。理论上, 这些更新表现为“脱离主成分”, 倾向于与预训练特征正交的低幅度方向 (Zhu 等人, 2025), 这与早期训练中观察到的秩-1 主导现象 (Cai 等人, 2025) 一致。然而, 这种模式面临着稳定性权衡的问题。KL-正则化 (“RL 的剃刀”) 试图限制遗忘 (Shenfeld 等人, 2025), 但可能导致“推理边界悖论”, 即激进的奖励最大化会压缩探索多样性 (Nguyen 等人, 2025)。尽管通过奇异向量旋转 (Jin 等人, 2025a,b) 可能部分逆转能力退化, 但适应仍然受到“隐形缰绳”的根本性限制, 该限制强制模型接近预训练流形 (Wu 等人, 2025)。从实证角度看, 仅更新一小部分参数即可获得显著提升, 这表明 RL 微调通常集中在小的子网络上 (Mukherjee 等人, 2025)。

2.2 PEFT and Spectral Priors

2.2 PEFT 和频谱先验

To alleviate the computational demands of scaling LLMs, PEFT (Xu et al., 2023; Han et al., 2024) has emerged as a key paradigm, minimizing memory overhead by updating only a fraction of parameters while matching full fine-tuning performance. Prevailing strategies include partial fine-tuning (Zaken et al., 2022; Lawton et al., 2023), soft prompt tuning (Hambardzumyan et al., 2021), non-linear adapters (Lin et al., 2020), low-rank adaptation (Li et al., 2018), and importance-aware methods like LIFT (Liu et al., 2025c). As a practical paradigm, PEFT has also been effectively adapted to continual learning (Liang et al., 2023) and multi-task learning (Liu et al., 2023) scenarios.

为了解决扩展大语言模型 (LLMs) 的计算需求, PEFT (Xu 等, 2023; Han 等, 2024) 作为一种关键范式应运而生, 通过仅更新部分参数来最小化内存开销, 同时达到全量微调的性能。主流策略包括部分微调 (Zaken 等, 2022; Lawton 等, 2023)、软提示调优 (Hambardzumyan 等, 2021)、非线性适配器 (Lin 等, 2020)、低秩适应 (Li 等, 2018) 以及像 LIFT (Liu 等, 2025c) 这样的基于重要性感知的方法。作为一种实用范式, PEFT 也被有效地应用于持续学习 (Liang 等, 2023) 和多任务学习 (Liu 等, 2023) 场景。

Among these, LoRA (Hu et al., 2021) and its variants are the de facto standard, yet their initialization strategies often diverge based on spectral priors. For instance, PiSSA (Meng et al., 2025) allocates trainable parameters to principal singular components, imposing a strong inductive bias validated primarily in SFT. However, such SFT-oriented spectral priors can create a fundamental geometric mismatch in RLVR, whose optimization dynamics and effective update patterns differ markedly from SFT (Yin et al., 2025; Zhu et al., 2025; Mukherjee et al., 2025). MiLoRA (Wang et al., 2025) instead targets minor components, but does not explicitly account for RLVR-specific geometry and dynamics. Other variants like DoRA (Liu et al., 2024), AdaLoRA (Zhang et al.,

其中, LoRA (Hu 等, 2021) 及其变体已成为事实上的标准, 但它们的初始化策略往往基于谱先验而有所不同。例如, PiSSA (Meng 等, 2025) 将可训练参数分配给主奇异成分, 施加了一个在 SFT 中主要验证的强归纳偏置。然而, 这种面向 SFT 的谱先验可能在 RLVR 中造成基本的几何不匹配, 因为 RLVR 的优化动态和有效的更新模式与 SFT 有显著差异 (Yin 等, 2025; Zhu 等, 2025; Mukherjee 等, 2025)。MiLoRA (Wang 等, 2025) 则针对次要成分, 但并未明确考虑 RLVR 特有的几何和动态特性。其他变体如 DoRA (Liu 等, 2024)、AdaLoRA (Zhang 等,

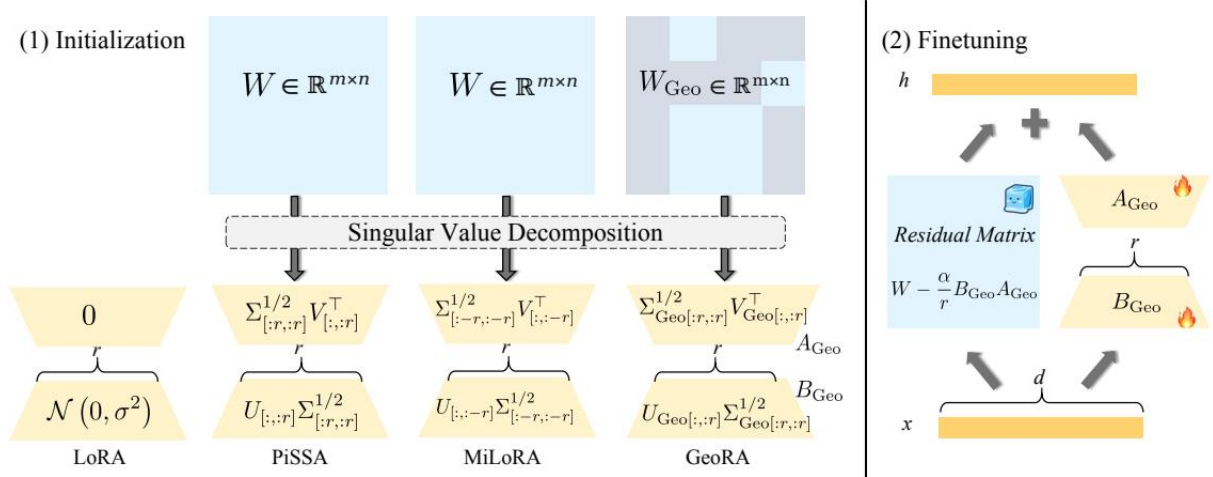


Figure 1: Comparison of adapter initialization and forward architectures. LoRA applies low-rank adaptation on the original weight matrix W with standard initialization, while PiSSA initializes adapters from the principal components of W . In contrast, GeoRA initializes from a geometry-constrained matrix W_{Geo} (a different adaptation target than W). Its forward pass incorporates a frozen Residual Matrix in parallel with the trainable adapter to act as a stability anchor for principal components.

图 1: 适配器初始化和前向架构的比较。LoRA 在原始权重矩阵 W 上应用标准初始化的低秩适应，而 PiSSA 则从 W 的主成分初始化适配器。相比之下，GeoRA 从一个几何约束矩阵 W_{Geo} （与 W 不同的适应目标）进行初始化。其前向传播过程中，将一个冻结的残差矩阵与可训练适配器并行作用，以作为主成分的稳定性锚点。

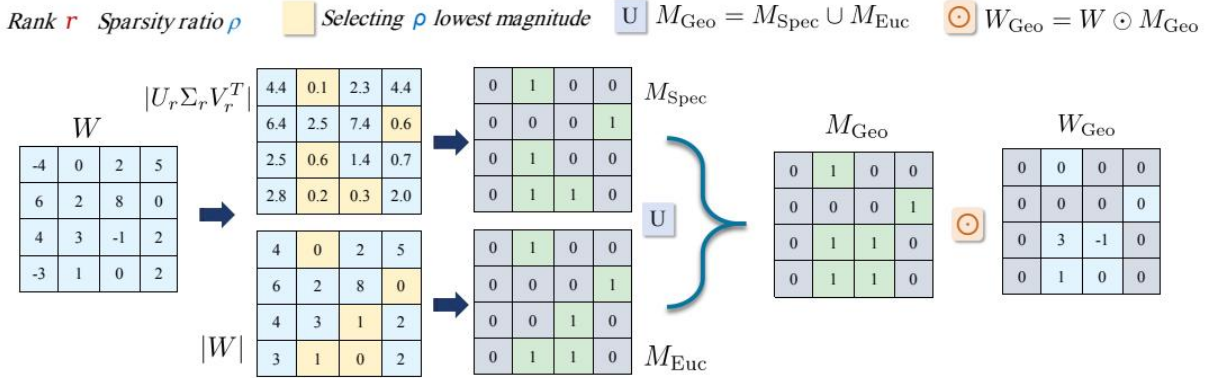


Figure 1: images/a029f501242905255fe7dd9871ae05468cd5d83e83a364e42175013f540ec1a8.jpg

Figure 2: Geometric Prior Construction via Masking. The process of generating M_{Geo} by combining Spectral Priors (low-curvature regions) and Euclidean Priors (high-plasticity near-zero weights). The resulting W_{Geo} isolates the most stable parameters for RL-native adaptation.

图 2: 通过掩码构建几何先验。生成 M_{Geo} 的过程是通过结合谱先验（低曲率区域）和欧几里得先验（接近零权重的高塑性区域）。最终得到的 W_{Geo} 会隔离出最适合用于原生强化学习适应的最稳定参数。

2023), and VeRA (Kopiczko et al., 2024) focus on weight decomposition or budget allocation without integrating these RLVR-native geometric considerations. Building on these observations, GeoRA translates RLVR-specific mechanistic insights into an actionable PEFT paradigm.

2023 年), 以及 VeRA (Kopiczko 等, 2024 年) 则专注于权重分解或预算分配, 而没有整合这些 RLVR 原生的几何考虑因素。基于这些观察, GeoRA 将 RLVR 特有的机制性见解转化为可操作的 PEFT 框架。

3 Methodology

3 方法论

We introduce GeoRA (Geometry-Aware Low-Rank Adaptation), a low-rank adaptation framework tailored for RLVR. GeoRA emphasizes an RL-native, structured low-rank parameterization together with an explicit residual leash for stability. Comparing with classical PEFT baselines like LoRA and PiSSA, we first present the low-rank forward architecture and initialization (Figure 1), and then describe a lightweight masking instantiation that constructs the geometry-constrained matrix used for initialization (Figure 2).

我们介绍 GeoRA (Geometry-Aware Low-Rank Adaptation)，这是一种针对 RLVR 的低秩适应框架。GeoRA 强调一种原生的 RL 结构化低秩参数化方法，以及用于稳定性的显式残差约束。与传统的 PEFT 基线方法如 LoRA 和 PiSSA 相比，我们首先介绍了低秩前向架构和初始化（图 1），然后描述了一种轻量级的掩码实现，用于构建用于初始化的几何约束矩阵（图 2）。

3.1 Geometry-Aware Low-Rank Structure

3.1 几何感知的低秩结构

Let W_{Geo} denote a geometry-constrained view of the pre-trained weight matrix W . Unlike standard LoRA which initializes adapters randomly (or with zero), GeoRA leverages the compressible structure within W_{Geo} to derive a structured initialization.

Let W_{Geo} denote a geometry-constrained view of the pre-trained weight matrix W . Unlike standard LoRA which initializes adapters randomly (or with zero), GeoRA leverages the compressible structure within W_{Geo} to derive a structured initialization.

We first perform Singular Value Decomposition (SVD) on the geometry-constrained matrix:

我们首先对几何约束矩阵进行奇异值分解 (SVD)：

$$W_{Geo} = U_{Geo} \Sigma_{Geo} V_{Geo}^T \quad (1)$$

We extract the top- r singular components that capture the principal geometric information. The low-rank adapters A_{Geo} and B_{Geo} are then initialized to approximate this geometry-aware subspace:

我们提取了捕捉主要几何信息的前 r 个奇异成分。然后将低秩适配器 A_{Geo} 和 B_{Geo} 初始化以近似这个几何感知的子空间：

$$A_{Geo} = \Sigma_{Geo[:, :r]}^{1/2} V_{Geo[:, :r]}^T \quad (2)$$

$$B_{Geo} = U_{Geo[:, :r]} \Sigma_{Geo[:, :r]}^{1/2} \quad (3)$$

By this design, the initial product $B_{Geo} A_{Geo}$ constructs the rank- r approximation of W_{Geo} .

通过这种设计，初始产品 $B_{Geo} A_{Geo}$ 构造了 W_{Geo} 的秩- r 近似。

Crucially, to ensure the model’s output remains unchanged at initialization and to preserve core capabilities during training, we follow the standard LoRA scaling and compute a Residual Matrix W_{res} by subtracting the scaled initialized adapters from the original weights:

关键的是，为了确保模型在初始化时的输出保持不变，并在训练过程中保留核心能力，我们遵循标准的 LoRA 缩放方法，并通过从原始权重中减去缩放后的初始化适配器来计算残差矩阵 W_{res} ：

$$W_{res} = W - \frac{\alpha}{r} B_{Geo} A_{Geo} \quad (4)$$

During the forward pass, W_{res} is kept frozen. The hidden state h is computed as:

在前向传播过程中， W_{res} 保持冻结。隐藏状态 h 的计算方式为：

$$h = \underbrace{W_{res}}_{\text{Frozen}} x + \underbrace{\frac{\alpha}{r} B_{Geo} A_{Geo}}_{\text{Trainable}} x \quad (5)$$

This construction keeps the model function-preserving at initialization (since $W_{res}x + \frac{\alpha}{r} B_{Geo} A_{Geo} x = Wx$) and enforces a hard structural constraint: the optimizer can only update the geometry-aligned manifold parameterized by A_{Geo} and B_{Geo} , while W_{res} acts as a stability anchor preventing the erosion of pre-trained representations.

这种构造在初始化时保持模型的功能不变（因为 $W_{res}x + \frac{\alpha}{r} B_{Geo} A_{Geo} x = Wx$ ），并强制执行一个严格的结构约束：优化器只能更新由 A_{Geo} 和 B_{Geo} 参数化几何对齐流形的参数，而 W_{res} 作为稳定性锚点，防止预训练表示的退化。

3.2 Geometric Prior Construction

3.2 几何先验构造

To instantiate a geometry-aware update region, we construct a geometry-constrained matrix W_{Geo} using a masking strategy. This masking is consistent with prior observations on spectral and magnitude structure and stability in fine-tuning (Liu et al., 2025c; Zhu et al., 2025).

为了实例化一个几何感知的更新区域，我们使用一种掩码策略构建一个几何约束矩阵 W_{Geo} 。这种掩码方式与之前在微调过程中对频谱和幅度结构及稳定性的观察结果一致（Liu 等，2025c；Zhu 等，2025）。

The Spectral Prior (M_{Spec}) promotes geometric stability by selecting the bottom ρ -fraction of entries from the rank- r approximation $\hat{W}_r$. The mask is defined as:

光谱先验 (M_{Spec}) 通过从秩为 r 的近似 $\hat{W}_r$ 中选择底部 ρ -分数的条目来促进几何稳定性。该掩码定义为：

$$(M_{Spec})_{i,j} = \mathbb{I}(|(\hat{W}_r)_{i,j}| \leq \tau_{Spec}(\rho)) \quad (6)$$

where $\tau_{Spec}(\rho)$ is the ρ -th quantile of the absolute values in $\hat{W}_r$. Intuitively, this mask suppresses high-magnitude (and typically high-curvature) components and constrains updates to a more stable, low-magnitude region, improving spectral stability under RLVR. Similarly, the Euclidean Prior (M_{Euc}) selects low-magnitude weights to capture parameter plasticity, using the same sparsity ratio ρ :

其中 $\tau_{Spec}(\rho)$ 是 ρ -th 分位数的绝对值在 $\hat{W}_r$ 中的值。直观来说，这个掩码会抑制高幅度（通常也是高曲率）的成分，并将更新限制在更稳定、低幅度的区域，从而在 RLVR 下提高频谱稳定性。同样地，欧几里得先验 (M_{Euc}) 选择低幅度的权重来捕捉参数可塑性，使用相同的稀疏率 ρ ：

$$(M_{\text{Euc}})_{i,j} = \mathbb{I}(|W_{i,j}| \leq \tau_{\text{Euc}}(\rho)) \quad (7)$$

Here, $\tau_{\text{Euc}}(\rho)$ represents the ρ -th quantile of $|W|$. The final geometry-constrained matrix W_{Geo} is

此处, $\tau_{\text{Euc}}(\rho)$ 表示 ρ -分位数的 $|W|$ 。最终的几何约束矩阵 W_{Geo} 是 formed by the union of these two stable subspaces:

由这两个稳定子空间的并集形成:

$$W_{\text{Geo}} = W \odot (M_{\text{Spec}} \cup M_{\text{Euc}}) \quad (8)$$

This union ensures that the optimized weights retain the flexibility of small parameters while respecting the spectral constraints of the pre-trained model.

这个联合确保了优化后的权重在保持小参数灵活性的同时,也尊重了预训练模型的频谱约束。

4 Experiments

4 实验

4.1 Experimental Setup

4.1 实验设置

We compare GeoRA against MiLoRA (Wang et al., 2025), PiSSA (Meng et al., 2025), LoRA (Hu et al., 2021), Sparse Fine-Tuning (SparseFT) (Zhu et al., 2025), and Full Fine-Tuning (FullFT). Our main experiments are conducted on mathematical RLVR, where we fine-tune Qwen3-8B-Base (Yang et al., 2025) and Llama-3.1-8B-Instruct (Grattafiori et al., 2024) on the DeepMath-103K dataset (He et al., 2025) using the GRPO algorithm (Shao et al., 2024), with a fixed rank $r = 16$ and sparsity ratio $\rho = 0.2$. We include both base and instruction-tuned backbones to evaluate robustness across model variants. In addition, we further validate our method on mathematical RLVR tasks with the 4B and 1.5B models on the GSM8K (Cobbe et al., 2021) dataset; details are provided in the Appendix C.

我们将 GeoRA 与 MiLoRA (Wang 等人, 2025)、PiSSA (Meng 等人, 2025)、LoRA (Hu 等人, 2021)、稀疏微调 (SparseFT) (Zhu 等人, 2025) 以及全微调 (FullFT) 进行比较。我们的主要实验是在数学 RLVR 上进行的, 我们在 DeepMath-103K 数据集 (He 等人, 2025) 上使用 GRPO 算法 (Shao 等人, 2024) 对 Qwen3-8B-Base (Yang 等人, 2025) 和 Llama-3.1-8B-Instruct (Grattafiori 等人, 2024) 进行微调, 固定秩为 $r = 16$, 稀疏率 $\rho = 0.2$ 。我们包括了基础模型和指令微调模型以评估不同模型变体的鲁棒性。此外, 我们在 GSM8K (Cobbe 等人, 2021) 数据集上使用 4B 和 1.5B 模型进一步验证了我们的方法在数学 RLVR 任务中的表现; 更多细节请参见附录 C。

For in-distribution (ID) evaluation, we assess mathematical reasoning performance on AIME24, AIME25 (Patel et al., 2024), MATH-500 (Hendrycks et al., 2021b), and Olym-MATH (Sun et al., 2025). For out-of-distribution (OOD) evaluation, we consider HumanEval (Coding) (Chen et al., 2021), GPQA (Science) (Rein et al., 2023), MMLU (General Knowledge) (Hendrycks et al., 2021a), IFEval (Instruction Following) (Zhou et al., 2023), and TruthfulQA (Truthfulness) (Lin et al., 2022) to measure cross-domain generalization and capability preservation. We also report the pre-RLVR Base model performance for direct comparison.

对于分布内 (ID) 评估, 我们在 AIME24、AIME25 (Patel 等人, 2024)、MATH-500 (Hendrycks 等人, 2021b) 和 Olym-MATH (Sun 等人, 2025) 上评估数学推理性能。对于分布外 (OOD) 评估, 我们考虑 HumanEval (编程) (Chen 等人, 2021)、GPQA (科学) (Rein 等人, 2023)、MMLU (常识) (Hendrycks 等人, 2021a)、IFEval (遵循指示) (Zhou 等人, 2023) 和 TruthfulQA (真实性) (Lin 等人, 2022), 以衡量跨领域泛化能力和能力保持。我们还报告了预 RLVR Base 模型的性能以进行直接比较。

Beyond the main mathematical setting, we further evaluate GeoRA on additional RLVR domains, including medical reasoning and coding. For medical RLVR, we train Llama-3.1-8B-Instruct on AlphaMed19K (Liu et al., 2025a) and report results on MedQA (Jin et al., 2021), MedMCQA (Pal et al., 2022), and PubMedQA (Jin et al., 2019). For coding, we train Qwen3-32B on Eurur-2-RL-Data (Cui et al., 2025) and evaluate on LiveCodeBench (Jain et al., 2025), HumanEval, and MBPP (Austin et al.,

在主要的数学设置之外, 我们进一步在额外的 RLVR 领域评估 GeoRA, 包括医疗推理和编码。对于医疗 RLVR, 我们在 AlphaMed19K (Liu 等, 2025a) 上训练 Llama-3.1-8B-Instruct, 并在 MedQA (Jin 等, 2021)、MedMCQA (Pal 等, 2022) 和 PubMedQA (Jin 等, 2019) 上报告结果。对于编码, 我们在 Eurur-2-RL-Data (Cui 等, 2025) 上训练 Qwen3-32B, 并在 LiveCodeBench (Jain 等, 2025)、HumanEval 和 MBPP (Austin 等,

Table 1: Main results on in-distribution (ID) mathematical RLVR benchmarks and out-of-distribution (OOD) tasks. Base denotes the original model before RLVR. Best results are in bold, and second-best results are underlined.

表 1: 在分布内 (ID) 数学 RLVR 基准测试和分布外 (OOD) 任务上的主要结果。Base 表示 RLVR 之前的原始模型。最佳结果加粗显示, 次佳结果下划线表示。

Method	In-Distribution (ID)				Out-of-Distribution (OOD)	
AIME24	AIME25	MATH500	OlymMATH	HumanEval	GPQA	MM
Qwen3-8B						
Base	13.33	11.67	71.20	9.75	76.83	36.9
FullFT	23.33	22.08	78.40	11.25	76.83	36.9
SparseFT	22.92	21.25	76.80	11.50	79.50	37.2
LoRA	19.58	19.58	75.60	10.75	81.10	37.5
PiSSA	22.50	20.42	74.40	11.75	71.95	36.1
MiLoRA	20.42	19.58	76.20	11.50	78.66	38.2
GeoRA	23.75	21.67	78.00	12.75	82.93	37.9
Llama-3.1-8B						
Base	9.58	2.08	51.00	3.25	65.20	31.1
FullFT	18.33	8.25	62.40	8.50	65.20	31.1
SparseFT	17.92	8.10	61.50	8.25	67.80	31.6
LoRA	15.42	6.25	58.20	6.75	69.80	32.1
PiSSA	17.50	7.92	60.50	7.75	67.50	31.8
MiLoRA	16.25	7.08	59.10	7.25	68.20	32.0
GeoRA	18.54	8.75	61.90	8.85	70.80	32.6

方法	In-Distribution (ID)				Out-of-Distribution (OOD)	
AIME24	AIME25	MATH500	OlymMATH	HumanEval	GPQA	MM
Qwen3-8B						

基础	13.33	11.67	71.20	9.75	76.83	36.9
全功能	23.33	22.08	78.40	11.25	76.83	36.9
稀疏 FT	22.92	21.25	76.80	11.50	79.50	37.2
LoRA	19.58	19.58	75.60	10.75	81.10	37.5
PiSSA	22.50	20.42	74.40	11.75	71.95	36.1
MiLoRA	20.42	19.58	76.20	11.50	78.66	38.2
GeoRA	23.75	21.67	78.00	12.75	82.93	37.9
Llama-3.1-8B						
基础	9.58	2.08	51.00	3.25	65.20	31.1
全职	18.33	8.25	62.40	8.50	65.20	31.1
稀疏 FT	17.92	8.10	61.50	8.25	67.80	31.6
LoRA	15.42	6.25	58.20	6.75	69.80	32.1
PiSSA	17.50	7.92	60.50	7.75	67.50	31.8
MiLoRA	16.25	7.08	59.10	7.25	68.20	32.0
GeoRA	18.54	8.75	61.90	8.85	70.80	32.6

Table 2: Results on additional medical and coding RLVR tasks. Best results are in bold, and second-best results are underlined.

表 2: 在额外的医学和编码 RLVR 任务上的结果。最佳结果以粗体表示，次佳结果以下划线表示。

Method	Medical				Coding			
MedQA	MedMCQA	PubMedQA	Average	LiveCodeBench	HumanEval	MBPP	Average	
Base	58.03	56.42	74.94	63.13	65.75	87.66	79.40	77.60
FullFT	75.32	64.56	80.12	73.33	67.75	90.26	81.40	79.80
LoRA	74.23	62.12	79.54	71.96	67.25	88.96	81.00	79.07
GeoRA	76.12	64.31	80.64	73.69	67.75	89.61	81.60	79.65

方法	医学				编程			
MedQA	MedMCQA	PubMedQA	平均值	LiveCodeBench	HumanEval	MBPP	平均值	
基础	58.03	56.42	74.94	63.13	65.75	87.66	79.40	77.60
全职	75.32	64.56	80.12	73.33	67.75	90.26	81.40	79.80
LoRA	74.23	62.12	79.54	71.96	67.25	88.96	81.00	79.07
GeoRA	76.12	64.31	80.64	73.69	67.75	89.61	81.60	79.65

2021). The implementation details and hyperparameter settings are provided in the Appendix C.

2021)。实施细节和超参数设置见附录 C。

4.2 Main Results

4.2 主要结果

Mathematical Reasoning Performance. Table 1 shows that GeoRA achieves the strongest overall performance on mathematical RLVR benchmarks across both backbones. It obtains the best results on most AIME and OlymMATH settings, while remaining competitive on MATH500. In particular, GeoRA consistently outperforms LoRA, MiLoRA, and PiSSA on challenging competition-style benchmarks, suggesting that geometry-aware initialization provides a more suitable inductive bias for RLVR than classic low-rank initialization.

数学推理表现。表 1 显示，GeoRA 在两种主干网络下，均在数学 RLVR 基准测试中取得了最强的整体表现。它在大多数 AIME 和 OlymMATH 设置中获得了最佳结果，同时在 MATH500 上仍保持竞争力。特别是，GeoRA 在具有挑战性的竞赛风格基准测试中持续优于 LoRA、MiLoRA 和 PiSSA，这表明几何感知的初始化为 RLVR 提供了比经典低秩初始化更合适的归纳偏差。

Out-of-Distribution Performance. GeoRA also generalizes favorably to OOD tasks. Across both backbones, it achieves the strongest or near-strongest results on HumanEval, GPQA, and MMLU, while generally retaining higher capability than other fine-tuning baselines on IFEval and

Out-of-Distribution 性能。GeoRA 也能够很好地泛化到 OOD 任务中。在两个主干网络上，它在 HumanEval、GPQA 和 MMLU 上取得了最强或接近最强的结果，而在 IFEval 上通常保持比其他微调基线更高的能力，并且在 IFEval 上通常保持比其他微调基线更高的能力。

TruthfulQA. These results indicate that constraining updates to geometry-aligned subspaces can improve in-domain reasoning performance while reducing interference with pre-existing general capabilities.

TruthfulQA。这些结果表明，将更新限制在与几何对齐的子空间中可以在提升领域内推理性能的同时，减少对已有通用能力的干扰。

Extension Beyond Mathematical RLVR. Beyond the main mathematical RLVR setting, we further evaluate GeoRA on medical and coding RLVR tasks. As shown in Table 2, GeoRA consistently outperforms LoRA across both domains and remains competitive with FullFT, suggesting that its advantage is not limited to mathematical reasoning.

扩展到数学以外的 RLVR 任务。除了主要的数学 RLVR 设置之外，我们还在医疗和编码 RLVR 任务上进一步评估 GeoRA。如表 2 所示，GeoRA 在两个领域内始终优于 LoRA，并且在与 FullFT 的竞争中保持竞争力，这表明其优势不仅限于数学推理。

4.3 Training Stability and Efficiency

4.3 训练的稳定性与效率

Training Dynamics. Figure 3 shows the training dynamics of Qwen3-8B as evaluated on the AIME benchmark. GeoRA remains consistently top-performing throughout training and reaches strong performance substantially earlier than other low-rank baselines. In contrast, LoRA, MiLoRA, and PiSSA improve more slowly and plateau at lower

训练动态。图 3 展示了在 AIME 基准上评估的 Qwen3-8B 的训练动态。GeoRA 在整个训练过程中始终保持领先表现，并且比其他低秩基线显著更早地达到强劲性能。相比之下，LoRA、MiLoRA 和 PiSSA 的提升速度较慢，并在较低水平上趋于平稳。

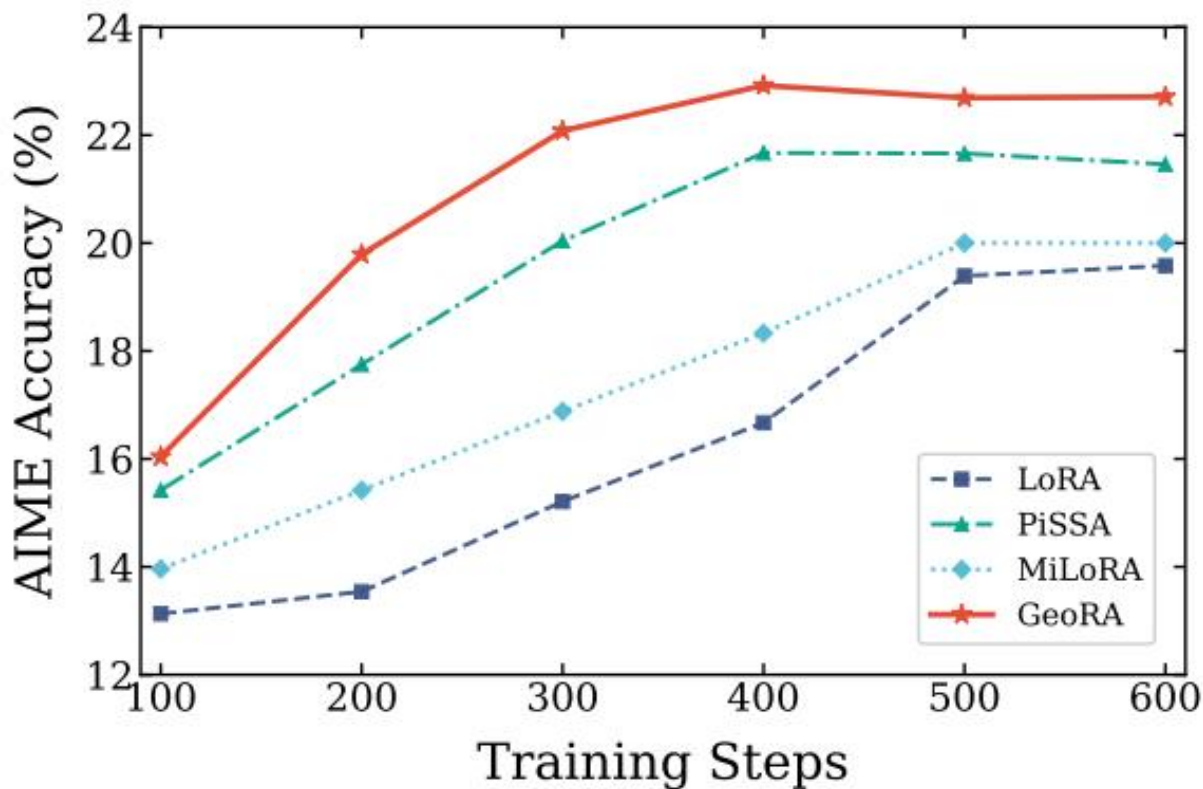


Figure 2: images/6d1384abd84035845a05e61e29678153717f8d38a9e811df4684a555b2cd4377.jpg

Figure 3: Training dynamics of Qwen3-8B as evaluated on the AIME benchmark (average of 2024 and 2025). GeoRA remains consistently top-performing throughout training.

图 3: Qwen3-8B 在 AIME 基准上的训练动态 (2024 和 2025 年的平均值)。在整个训练过程中, GeoRA 始终保持领先表现。

levels. This indicates that GeoRA not only yields better final accuracy, but also provides a more favorable optimization trajectory under RLVR.

级别。这表明 GeoRA 不仅能够获得更好的最终精度, 而且在 RLVR 下提供了更有利的优化轨迹。

Hyperparameter Robustness. We first evaluate performance under different learning rates on Qwen3-4B in Figure 4. GeoRA maintains high reward across a broad range of learning rates, while other low-rank baselines are much more sensitive to this choice. In particular, PiSSA and MiLoRA degrade rapidly under larger learning rates, and LoRA also exhibits a clear drop in performance at the high end of the sweep. These results suggest that GeoRA is more robust to hyperparameter variation and requires less delicate tuning in practice. Similar robustness trends are also observed for other hyperparameters, including rank and sparsity, as shown in Appendix D.

超参数鲁棒性。我们首先在图 4 中评估 Qwen3-4B 在不同学习率下的性能。GeoRA 在广泛的学习率范围内保持较高的奖励, 而其他低秩基线则对该选择更加敏感。特别是, PiSSA 和 MiLoRA 在较大的学习率下迅速下降, LoRA 在扫描范围的高端也表现出明显的性能下降。这些结果表明, GeoRA 对超参数变化更具鲁棒性, 并且在实践中需要更少的精细调优。其他超参数, 包括秩和稀疏度, 也观察到了类似的鲁棒性趋势, 如附录 D 所示。

Stability Under Aggressive Optimization. To stress-test optimization stability, we further compare reward and KL divergence under an aggressive learning rate in Figure 5. GeoRA maintains the highest reward trajectory without collapse, whereas PiSSA suffers a catastrophic drop late in training. At the same time, GeoRA keeps KL divergence at a low and controlled level throughout

training. These results are highly relevant to RLVR optimization, which often requires stable policy updates within trust-region-like boundaries. The combination of higher reward and smoother KL behavior suggests that GeoRA improves exploration and policy refinement without inducing the destabilizing updates caused by geometry-misaligned adaptation.

在激进优化下的稳定性。为了测试优化的稳定性，我们在图 5 中进一步比较了在激进学习率下的奖励和 KL 散度。GeoRA 保持了最高的奖励轨迹而没有崩溃，而 PiSSA 在训练后期经历了灾难性的下降。同时，GeoRA 在整个训练过程中保持 KL 散度处于较低且可控的水平。这些结果与 RLVR 优化高度相关，因为通常需要在类似信任区域的边界内保持稳定的策略更新。较高的奖励和更平滑的 KL 行为的结合表明，GeoRA 在不引起由几何不匹配适应导致的不稳定更新的情况下，提高了探索能力和策略优化。

Efficiency. Table 3 compares the training efficiency of FullFT, SparseFT, and GeoRA. GeoRA updates only a tiny fraction of parameters, reducing trainable parameters by 99.5% relative to FullFT, while also lowering VRAM usage and improving per-iteration training speed. Compared with SparseFT, GeoRA avoids the overhead of unstructured sparse computation and therefore translates its parameter efficiency into actual hardware efficiency. Although GeoRA introduces an SVD-based initialization step, this cost is a one-time preprocessing overhead and is negligible compared with RLVR training; detailed profiling is provided in the appendix. Overall, these results show that GeoRA improves not only effectiveness and stability, but also practical training efficiency.

效率。表 3 比较了 FullFT、SparseFT 和 GeoRA 的训练效率。GeoRA 仅更新极小一部分参数，相较于 FullFT，可训练参数减少了 99.5%，同时降低了 VRAM 的使用并提高了每轮训练的速度。与 SparseFT 相比，GeoRA 避免了非结构化稀疏计算的开销，因此将参数效率转化为实际的硬件效率。虽然 GeoRA 引入了一个基于 SVD 的初始化步骤，但这一成本仅是一次性预处理开销，并且与 RLVR 训练相比可以忽略不计；详细的性能分析见附录。总体而言，这些结果表明，GeoRA 不仅提高了效果和稳定性，还提升了实际的训练效率。

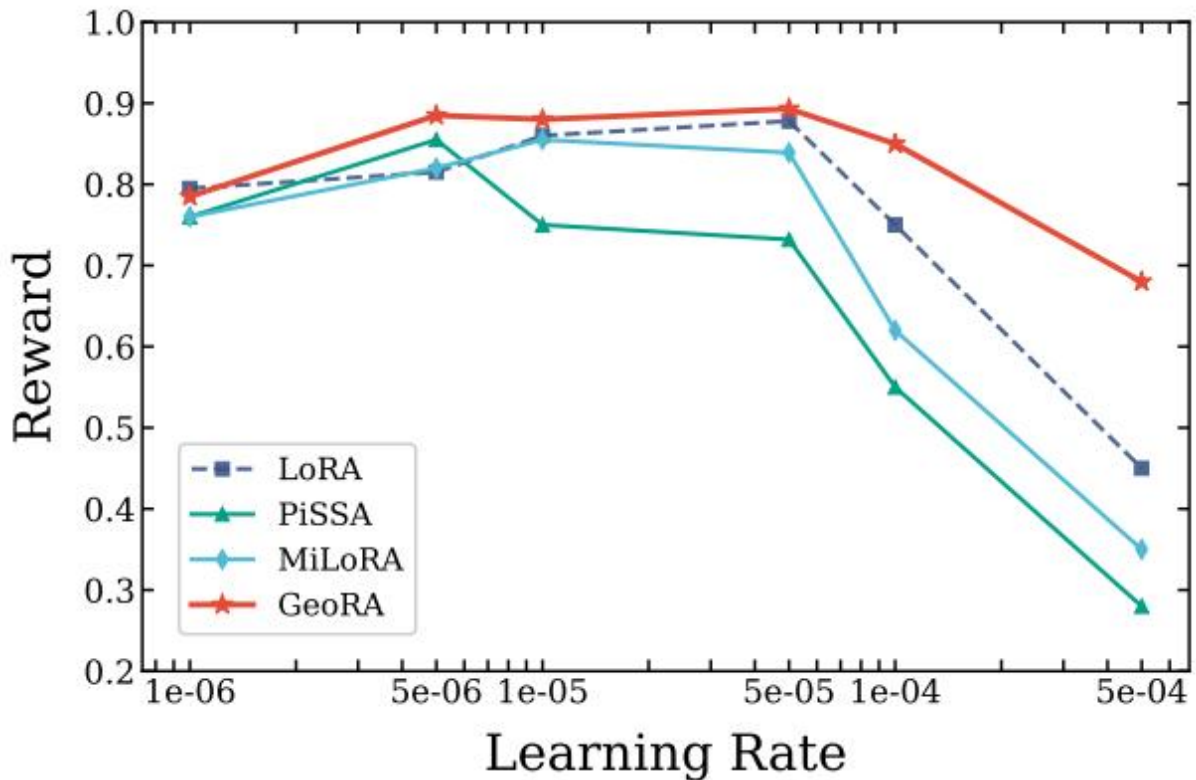


Figure 3: images/00e3689068e09a784d15ddf05741148ffc8236f61802e07bd9e85d459e05f57a.jpg

Figure 4: Performance comparison across different learning rates. GeoRA demonstrates superior

stability and robust convergence even at higher learning rates.

图 4: 不同学习率下的性能比较。GeoRA 在较高的学习率下也表现出优越的稳定性与鲁棒的收敛性。

Table 3: Efficiency Comparison between Full FT, SparseFT, and GeoRA. Relative reductions compared to Full FT are shown in parentheses.

表 3: 全量 FT、稀疏 FT 和 GeoRA 的效率对比。与全量 FT 相比的相对降幅用括号表示。

Method	Params (B)	Time (s/it)	VRAM (%)
Full FT	8.00	231	95.73
SparseFT	2.56(-68.0%)	256(+10.8%)	81.25(-15.1%)
GeoRA	0.04(-99.5%)	185(-19.9%)	68.43(-28.5%)

方法	参数 (B)	时间 (s/it)	VRAM (%)
完整 FT	8.00	231	95.73
稀疏 FT	2.56(-68.0%)	256(+10.8%)	81.25(-15.1%)
GeoRA	0.04(-99.5%)	185(-19.9%)	68.43(-28.5%)

4.4 Ablation and Analysis

4.4 去除实验与分析

Impact of Initialization Strategy. Table 4 shows that GeoRA’s geometry-aware initialization is critical to its performance. Replacing the proposed initialization with Random- r initialization leads to a clear drop across all benchmarks, while Tail- r initialization performs even worse. This indicates that the benefit of GeoRA does not come from low-rank adaptation alone, but from aligning the initialization with the effective RLVR update subspace. In particular, the poor performance of Tail- r suggests that simply selecting low-energy

初始化策略的影响。表 4 显示，GeoRA 的几何感知初始化对其性能至关重要。将提出的初始化替换为 Random- r 初始化会导致所有基准测试中的表现明显下降，而 Tail- r 初始化的表现甚至更差。这表明 GeoRA 的优势不仅仅来自于低秩适应，而是来自于将初始化与有效的 RLVR 更新子空间对齐。特别是，Tail- r 表现不佳表明，仅仅选择低能量的

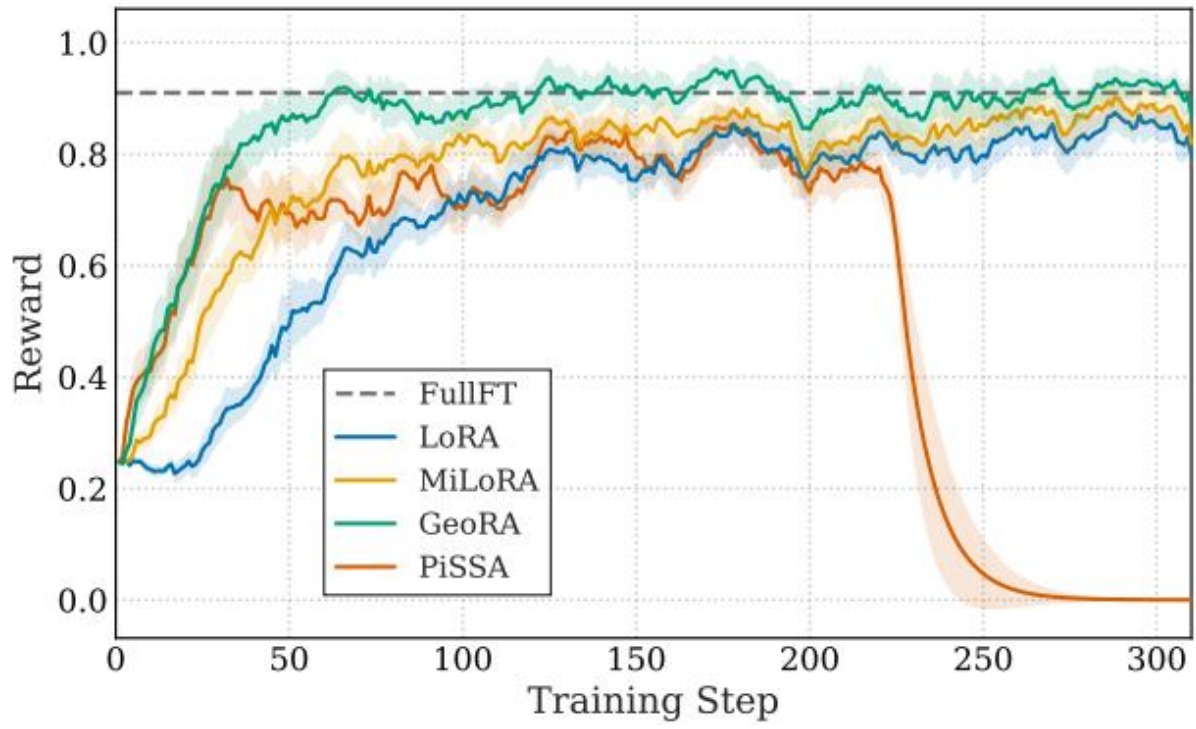


Figure 4: <images/e738eee072b7700cae20f01657e85d42c16b3fb179db493475775793bc1bfbcf.jpg>

(a) Reward Trajectories

(a) 奖励轨迹

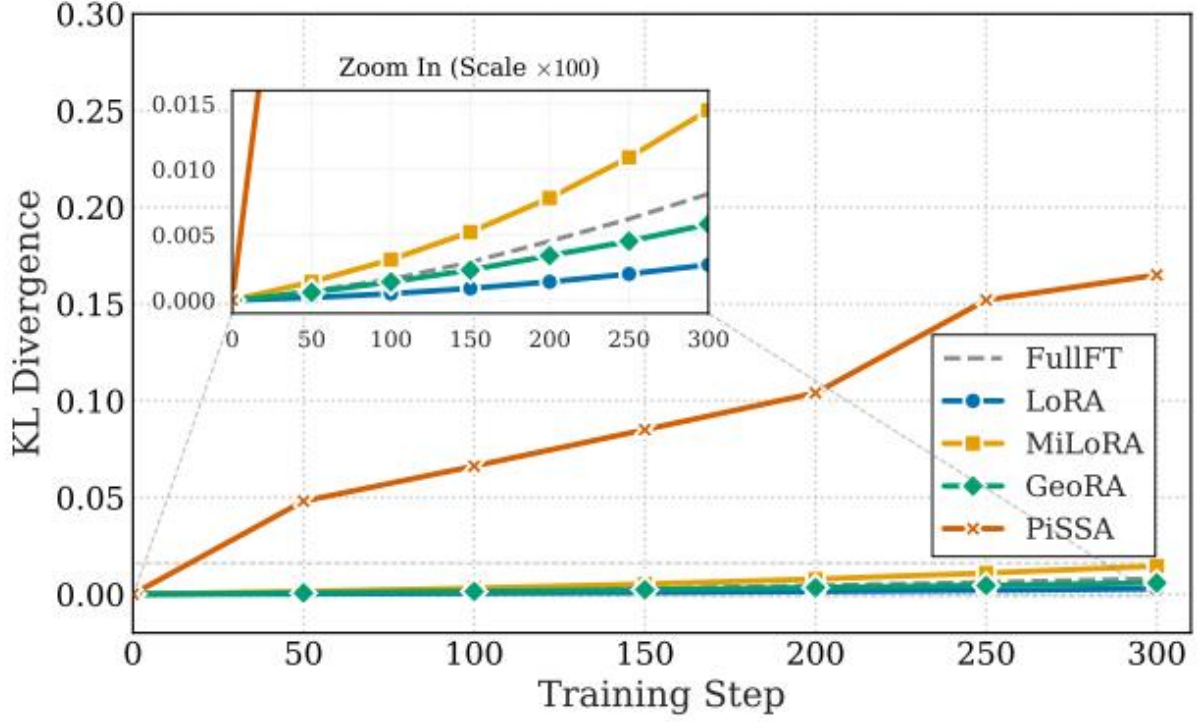


Figure 5: images/cd3f654c6d6ba5c53d491ac537038db979dc7af6651858c6ad56a4cc6a553847.jpg

(b) KL Divergence

(b) KL 散度

Figure 5: Training Stability and Constraint Adherence. Results on Qwen3-4B show that GeoRA demonstrates superior robustness under aggressive learning rates (5×10^{-5}).

图 5: 训练稳定性与约束遵循性。在 Qwen3-4B 上的结果表明, GeoRA 在激进的学习率下表现出更强的鲁棒性 (5×10^{-5})。

Method	Reward	AIME24	AIME25	MATH-500	OlymMATH	Average
GeoRA	0.88	13.33	9.17	73.40	5.75	25.41
Ablation on Initialization Strategy						
Random-r Init	0.85	12.50	8.50	72.10	5.25	24.60
Tail-r Init	0.82	11.67	7.50	70.80	4.50	23.40
Ablation on Geometric Masks						
w/o M_{Spec}	0.86	12.50	8.33	72.00	4.75	24.40
w/o M_{Euc}	0.83	13.33	8.75	72.80	5.50	25.10

方法	奖励	AIME24	AIME25	MATH-500	OlymMATH	平均值
GeoRA	0.88	13.33	9.17	73.40	5.75	25.41
初始化策略的消融实验						
随机初始化	0.85	12.50	8.50	72.10	5.25	24.60

尾部初始化	0.82	11.67	7.50	70.80	4.50	23.40
几何掩码的消融实验						
w/o M_{Spec}	0.86	12.50	8.33	72.00	4.75	24.40
无 M_{Euc}	0.83	13.33	8.75	72.80	5.50	25.10

Table 4: Ablation study on Qwen3-4B with different initialization strategies and geometric mask variants.

表 4：在不同初始化策略和几何掩码变体下对 Qwen3-4B 进行消融研究。

directions is insufficient; the trainable directions must still capture the dominant structure within the geometry-constrained subspace.

方向性不足；可训练的方向性仍必须捕捉几何约束子空间内的主导结构。

Impact of Geometric Masks. We further ablate the two mask components used to construct the geometry-constrained subspace. Removing either M_{Spec} or M_{Euc} consistently degrades performance, confirming that both are necessary for strong results. This suggests that the two priors play complementary roles: the spectral mask helps suppress unstable high-energy directions, while the Euclidean mask preserves adaptation flexibility in low-magnitude regions. Their combination therefore provides a better approximation to the effective RLVR update manifold than either component alone.

几何掩码的影响。我们进一步消融用于构建几何约束子空间的两个掩码组件。移除任一 M_{Spec} 或 M_{Euc} 都会持续降低性能，这证实了两者对于获得良好结果都是必要的。这表明这两个先验知识在功能上是互补的：谱掩码有助于抑制不稳定高能量方向，而欧几里得掩码则在低幅值区域保持适应的灵活性。因此，它们的组合比单独使用任何一个组件更能提供对有效 RLVR 更新流形更好的近似。

Overall Analysis. Taken together, the ablation results show that GeoRA’s gains arise from the full geometry-aware design rather than from any single isolated choice. The geometry-constrained subspace, structured initialization, and frozen residual anchor work together to improve optimization quality while preserving pre-trained structure. This is consistent with the stability and efficiency results in the previous section, and further supports

总体分析。综合来看，消融实验的结果表明，GeoRA 的性能提升来自于完整的几何感知设计，而不是任何单一的独立选择。几何约束子空间、结构化初始化和冻结残差锚点共同作用，提高了优化质量，同时保持了预训练结构。这与前一节中的稳定性和效率结果是一致的，并进一步支持了

our claim that RLVR benefits from a low-rank parameterization explicitly aligned with its update geometry.

我们主张 RLVR 从一种与更新几何明确对齐的低秩参数化中获益。

5 Mechanism Analysis

5 机制分析

5.1 Geometric Priors

5.1 几何先验

GeoRA constructs its update subspace from two geometric priors: a spectral prior and a Euclidean prior. The motivation is that RLVR updates are highly selective, tending to exploit under-utilized yet plastic regions of the pre-trained model while avoiding large modifications to dominant principal directions. Accordingly, M_{Spec} suppresses high-energy components in the principal subspace, whereas M_{Euc} selects low-magnitude parameters in the original weight space. Their union defines a subspace that is both stable and expressive for RLVR adaptation.

GeoRA 从两个几何先验构建其更新子空间：一个谱先验和一个欧几里得先验。其动机在于 RLVR 更新具有高度选择性，倾向于利用预训练模型中未充分利用但具有可塑性的区域，同时避免对主导主方向进行大幅修改。因此， M_{Spec} 抑制主子空间中的高能量成分，而 M_{Euc} 在原始权重空间中选择低幅度参数。它们的联合定义了一个既稳定又具有表达能力的子空间，用于 RLVR 的适应。

We further verify that these two priors are complementary rather than redundant in Table 5. On Qwen3-8B with $\rho = 0.2$, both M_{Spec} and M_{Euc} select 20.0% of parameters, but their overlap remains small: the overall intersection is only 4.55%, with a Jaccard index of 0.128. Similar patterns are observed in both MLP layers (4.63%, 0.131)

我们进一步验证了这两个先验在表 5 中是互补而非冗余的。在 Qwen3-8B 中， $\rho = 0.2$ ， M_{Spec} 和 M_{Euc} 都选择了 20.0% 的参数，但它们的重叠仍然很小：整体交集仅为 4.55%，Jaccard 指数为 0.128。在两个 MLP 层中也观察到了类似的模式（4.63%，0.131）

Table 5: Overlap analysis of the spectral prior M_{Spec} and Euclidean prior M_{Euc} on Qwen3-8B with $\rho = 0.2$.

表 5: Qwen3-8B 中光谱先验 M_{Spec} 与欧几里得先验 M_{Euc} 的重叠分析 $\rho = 0.2$ 。

Group	M_{Spec}	M_{Euc}	Intersection	Jaccard
MLP	20.0%	20.0%	4.63%	0.131
Attn	20.0%	20.0%	4.25%	0.119
All	20.0%	20.0%	4.55%	0.128

组	M_{Spec}	M_{Euc}	交叉路口	Jaccard
多层感知器 (Multilayer Perceptron)	20.0%	20.0%	4.63%	0.131
注意	20.0%	20.0%	4.25%	0.119
全部	20.0%	20.0%	4.55%	0.128

and attention layers (4.25%, 0.119). This limited overlap indicates that the two masks capture largely distinct parameter subsets and therefore serve complementary geometric roles. This is also consistent with the ablation results in Table 4: removing either prior leads to a clear performance drop.

以及注意力层 (4.25%, 0.119)。这种有限的重叠表明, 这两个掩码捕捉的是基本上不同的参数子集, 因此在此在几何作用上具有互补性。这也与表 4 中的消融实验结果一致: 移除任一先验会导致性能明显下降。

5.2 Low-Rank Structure

5.2 低秩结构

To validate the structural basis of GeoRA, we analyze the singular value spectrum of the geometry-constrained subspace. We compare W_{Geo} with the original pre-trained weights, as well as random noise matrices at both full and sparse density. Figure 6a yields two key insights. First, sparsity does not by itself induce low-rankness. The spectrum of sparse random noise is nearly indistinguishable from that of dense random noise, and both follow a relatively flat decay pattern. This indicates that unstructured sparsity alone remains essentially isotropic in the spectral domain. Second, W_{Geo} preserves a pronounced heavy-tailed spectrum similar to that of the pre-trained weights, with most spectral mass concentrated in a small number of leading components. This shows that the selected update region inherits a structured and compressible low-rank form rather than behaving like random sparse noise.

为了验证 GeoRA 的结构基础, 我们分析几何约束子空间的奇异值谱。我们将 W_{Geo} 与原始预训练权重以及在全密度和稀疏密度下的随机噪声矩阵进行比较。图 6a 得出了两个关键见解。首先, 稀疏性本身并不能诱导低秩性。稀疏随机噪声的谱与密集随机噪声的谱几乎无法区分, 且两者都呈现出相对平坦的衰减模式。这表明, 无结构的稀疏性在谱域中仍然基本上是各向同性的。其次, W_{Geo} 保留了一个明显的重尾谱, 与预训练权重的谱相似, 大部分谱质量集中在少量的主成分上。这表明所选的更新区域继承了一种结构化且可压缩的低秩形式, 而不是像随机稀疏噪声那样行为。

We further find that the actual FullFT update, $\Delta W_{FullFT} = W_{FullFT} - W_{Pretrain}$, exhibits a similarly compressible heavy-tailed spectrum. This strongly supports the low-rank inductive bias of GeoRA: the effective RLVR update space is not isotropic, but already structured and highly compressible. GeoRA therefore captures an intrinsic property of RLVR updates, instead of imposing an artificial low-rank constraint.

我们进一步发现, 实际的 FullFT 更新 $\Delta W_{FullFT} = W_{FullFT} - W_{Pretrain}$ 也表现出类似的可压缩的重尾谱。这强烈支持了 GeoRA 的低秩归纳偏见: 有效的 RLVR 更新空间并不是各向同性的, 而是已经具有结构并且高度可压缩的。因此, GeoRA 捕捉了 RLVR 更新的内在属性, 而不是强加一个人为的低秩约束。

5.3 Spectral Efficiency and Alignment

5.3 频谱效率与对齐

Building on the low-rank structure of RL updates, we validate GeoRA’s geometric superiority via Normalized Spectral Shift (NSS) and Subspace Alignment (Table 6). First, NSS quantifies the topologi-

基于 RL 更新的低秩结构, 我们通过归一化谱位移 (NSS) 和子空间对齐 (表 6) 验证了 GeoRA 的几何优势。首先, NSS 量化了拓扑

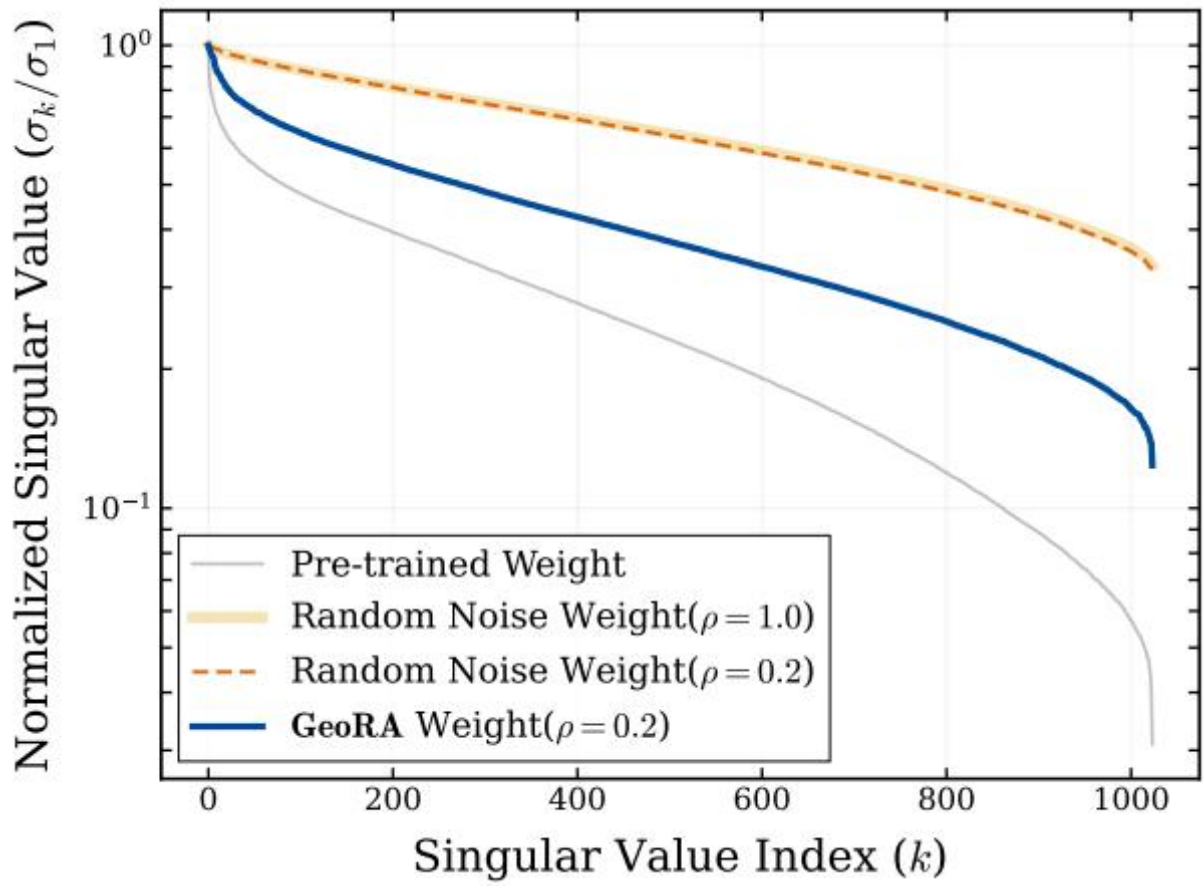


Figure 6: images/3057d6756d21e4449d8519690d3999e6a67bb080d963281cf782e2caca41751c.jpg

(a) Singular value spectrum of the geometry-constrained subspace W_{Geo} .

(a) 几何约束子空间 W_{Geo} 的奇异值谱。

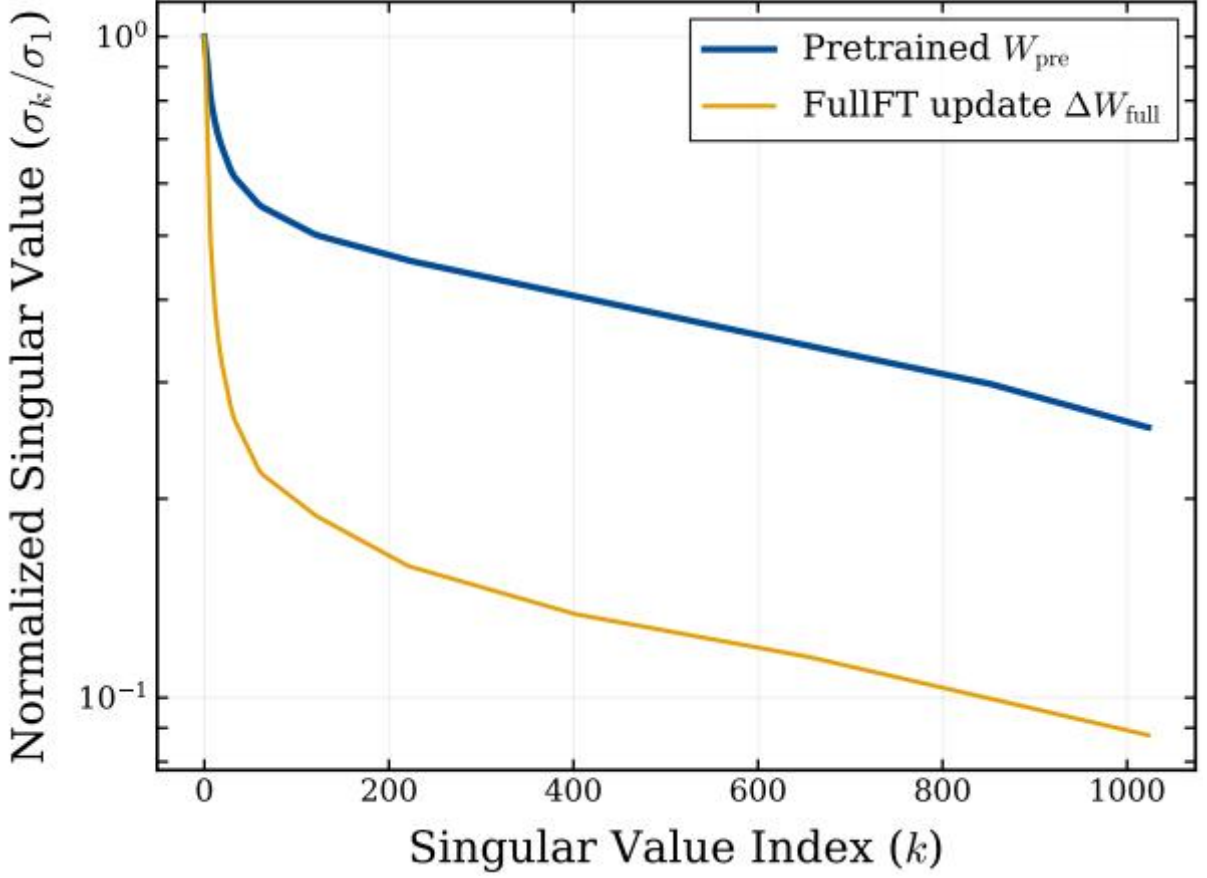


Figure 7: images/18498dc11ad44ba6c462de443612816deff17ed3d7bf243870b0642b76351ebc.jpg

(b) Singular value spectrum of the full fine-tuning update ΔW_{FullFT} .
Figure 6: Spectral analysis of RLVR update structure.

(b) 全局微调更新 ΔW_{FullFT} 的奇异值谱。
图 6: RLVR 更新结构的谱分析。

cal distortion of the pre-trained manifold:

预训练流形的校正失真:

$$NSS = \frac{\|\sigma(W^{\text{tuned}}) - \sigma(W)\|_2}{\|\sigma(W)\|_2}. \quad (9)$$

While PiSSA exhibits high NSS (> 0.39), indicating aggressive structural modification, GeoRA maintains minimal distortion (≈ 0.09), confirming it maximizes reward acquisition while preserving fundamental features.

虽然 PiSSA 表现出较高的 NSS (> 0.39), 表明其具有强烈的结构修改特性, 但 GeoRA 保持了最小的失真 (≈ 0.09), 证实了它在获取奖励的同时最大限度地保留了基本特征。

To pinpoint the update locus, we calculate the alignment between ΔW and the pre-trained singular vectors V :

为了确定更新的位置, 我们计算 ΔW 与预训练的单个向量 V 之间的对齐情况:

$$\mathcal{S}(k) = \frac{\|\Delta W v_k\|_2}{\|\Delta W\|_F} \approx |\cos(\theta_k)|. \quad (10)$$

GeoRA shows a distinct signature: it avoids the high-energy principal subspace ($S_{\text{Head}} \leq 0.02$)—enhancing stability—and efficiently adapts the geometry-constrained tail ($S_{\text{Tail}} \geq 0.96$). In contrast, PiSSA’s instability is explained by its high overlap with the head subspace (≈ 0.98).

GeoRA 显示了一个明显的特征：它避免了高能主子空间 ($S_{\text{Head}} \leq 0.02$) ——从而增强稳定性，并高效地适应了几何约束的尾部 ($S_{\text{Tail}} \geq 0.96$)。相比之下，PiSSA 的不稳定性是由于其与头部子空间 (≈ 0.98) 的高度重叠所致。

Table 6: Geometric mechanism analysis.

表 6：几何机构分析。

	Llama-3.1-8B			Qwen3-8B		
Method	NSS□	$\mathcal{S}_H \downarrow$	$\mathcal{S}_T \uparrow$	NSS□	$\mathcal{S}_H \downarrow$	$\mathcal{S}_T \uparrow$
PiSSA	0.395	0.98	0.01	0.418	0.95	0.03
LoRA	0.214	0.15	0.15	0.235	0.18	0.18
MiLoRA	0.125	0.12	0.92	0.132	0.16	0.90
GeoRA	0.092	0.005	0.98	0.096	0.015	0.96

	Llama-3.1-8B			Qwen3-8B		
方法	NSS□	$\mathcal{S}_H \downarrow$	$\mathcal{S}_T \uparrow$	NSS□	$\mathcal{S}_H \downarrow$	$\mathcal{S}_T \uparrow$
PiSSA	0.395	0.98	0.01	0.418	0.95	0.03
LoRA	0.214	0.15	0.15	0.235	0.18	0.18
MiLoRA	0.125	0.12	0.92	0.132	0.16	0.90
GeoRA	0.092	0.005	0.98	0.096	0.015	0.96

6 Conclusion

6 结论

In this paper, we bridge the gap between parameter-efficient fine-tuning and the distinctive optimization geometry of RLVR. Our study suggests that existing PEFT methods often yield suboptimal results in RLVR due to a structural mismatch. SFT-oriented low-rank methods prioritize directions that may be misaligned with the geometry-constrained updates favored by RLVR, while some sparse update methods better reflect RLVR dynamics but fail to translate into practical hardware efficiency. GeoRA addresses these challenges by identifying a compressible, geometry-aligned update subspace and parameterizing it through structured SVD initialization with a frozen residual anchor. Experiments on Qwen and Llama models from 1.5B to 32B parameters show that GeoRA consistently improves mathematical RLVR performance, extends effectively to medical and coding RLVR settings, and exhibits stronger generalization with less forgetting on out-of-domain tasks. Overall, GeoRA provides an effective, stable, and hardware-efficient adaptation strategy for RLVR.

在本文中，我们弥合了参数高效微调与 RLVR 独特优化几何之间的差距。我们的研究表明，现有的 PEFT 方法由于结构不匹配，常常在 RLVR 中产生次优结果。面向 SFT 的低秩方法优先考虑的方向可能与 RLVR 偏好的几何约束更新不一致，而一些稀疏更新方法更能反映 RLVR 的动力学特性，但无法转化为实际的硬件效率。GeoRA 通过识别一个可压缩且与几何对齐的更新子空间，并通过冻结残差锚点的结构化 SVD 初始化对其进行参数化，从而解决了这些挑战。在 Qwen 和 Llama 模型（参数量从 1.5B 到 32B）上的实验表明，GeoRA 在数学 RLVR 任务中持续提升性能，能够有效扩展到医疗和编码 RLVR 场景，并在领域外任务中表现出更强的泛化能力且遗忘更少。总体而言，GeoRA 为 RLVR 提供了一种有效、稳定且硬件高效的适应策略。

Limitations

限制

Despite its effectiveness, GeoRA has certain limitations. First, the initialization process requires performing a truncated SVD and dual-masking operations. While these represent a one-time computational cost at the beginning of training, they introduce an additional pre-processing step compared to the random initialization used in standard LoRA. Second, our experiments primarily focus on RLVR in reasoning domains. Although GeoRA demonstrates strong performance on these tasks, its generalizability needs to be further validated across a broader range of model architectures and diverse reinforcement learning scenarios beyond verifiable rewards.

尽管 GeoRA 具有良好的效果，但它仍存在一些限制。首先，初始化过程需要执行截断 SVD 和双掩码操作。虽然这些操作在训练开始时代表了一次性的计算成本，但与标准 LoRA 中使用的随机初始化相比，它们引入了额外的预处理步骤。其次，我们的实验主要集中在推理领域中的 RLVR 任务上。尽管 GeoRA 在这些任务上表现出色，但其泛化能力仍需在更广泛的模型架构和多样化的强化学习场景（超越可验证奖励）中进一步验证。

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